Laser

Learning Objective: Learners will be able to…

(1) Solve problems on absorption, spontaneous, and stimulated emission.

(2) Understand operation of resonant cavity.

(3) Understand He-Ne laser and application.

(4) Describe how three dimensional image is recorded on hologram and reconstructed.

LASER:

Laser is an acronym for “Light Amplification by Stimulated Emission of Radiation”

It is a source which emits an intense, almost perfectly monochromatic, directional and highly coherent beam of light.

Characteristics of Laser Light

1. Coherence 2. Highly Monochromatic 3. Highly Directional and Collimated

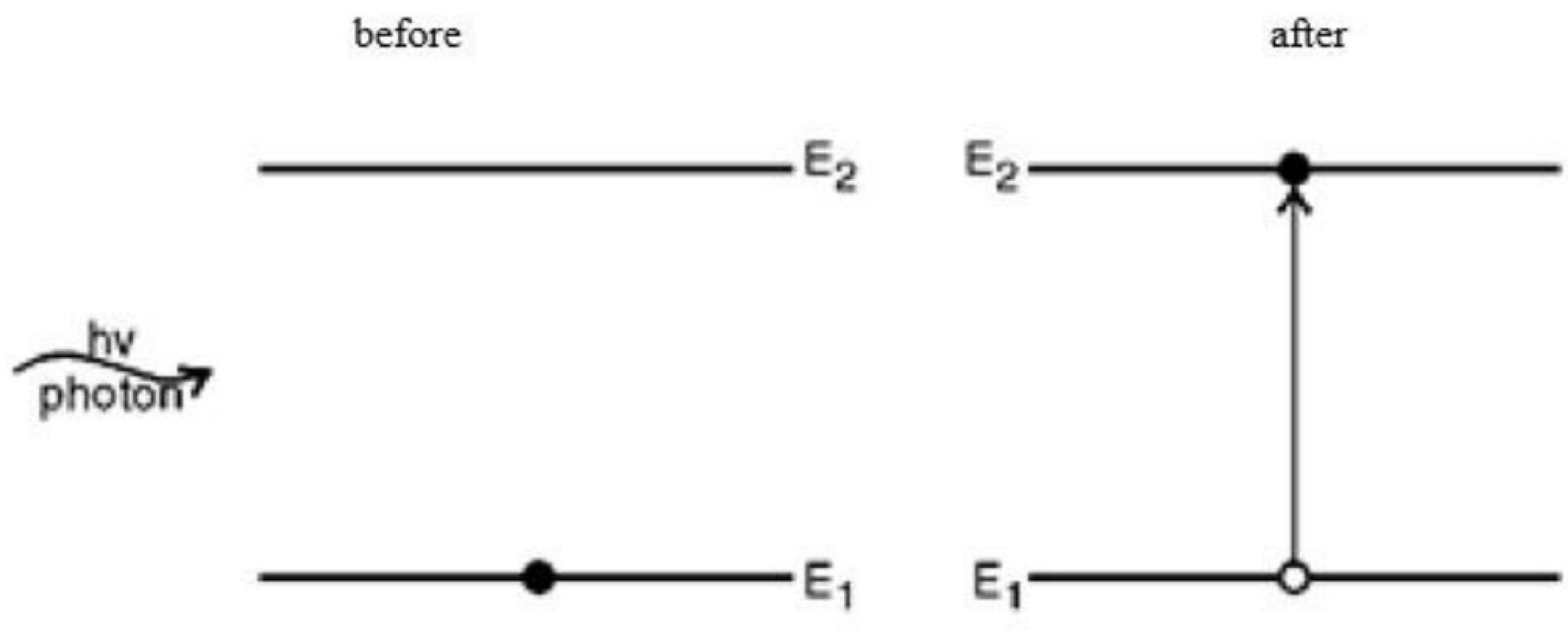
4. Polarized

5. Highly Energetic

6. Negligible divergence

Interaction of radiation with matter: In atomic system, atoms occupy the lowest energy state, known as ground state. An atom in a lower energy state may be excited to a higher energy state by absorption of electromagnetic radiation of suitable frequency. Einstein considered following processes.

(1)Induced absorption: In atomic system atom has discrete energy states, the atoms occupy the lowest energy state known as ground state. The atom in a lower energy state can be raised to higher energy state by absorbing a photon of frequency v .This is stimulated absorption or induced absorption.



Induced absorption

The process may be expressed as

$$A + hv = A^*$$

Where A is an atom in the lower energy state E_1 and A^* is an exited atom in higher energy state E_2

Let N_1 and N_2 number of atoms be the per unit volume in energy state E_1 and E_2.

The number of absorption transition per unit volume per unit time or probability rate of absorption transition

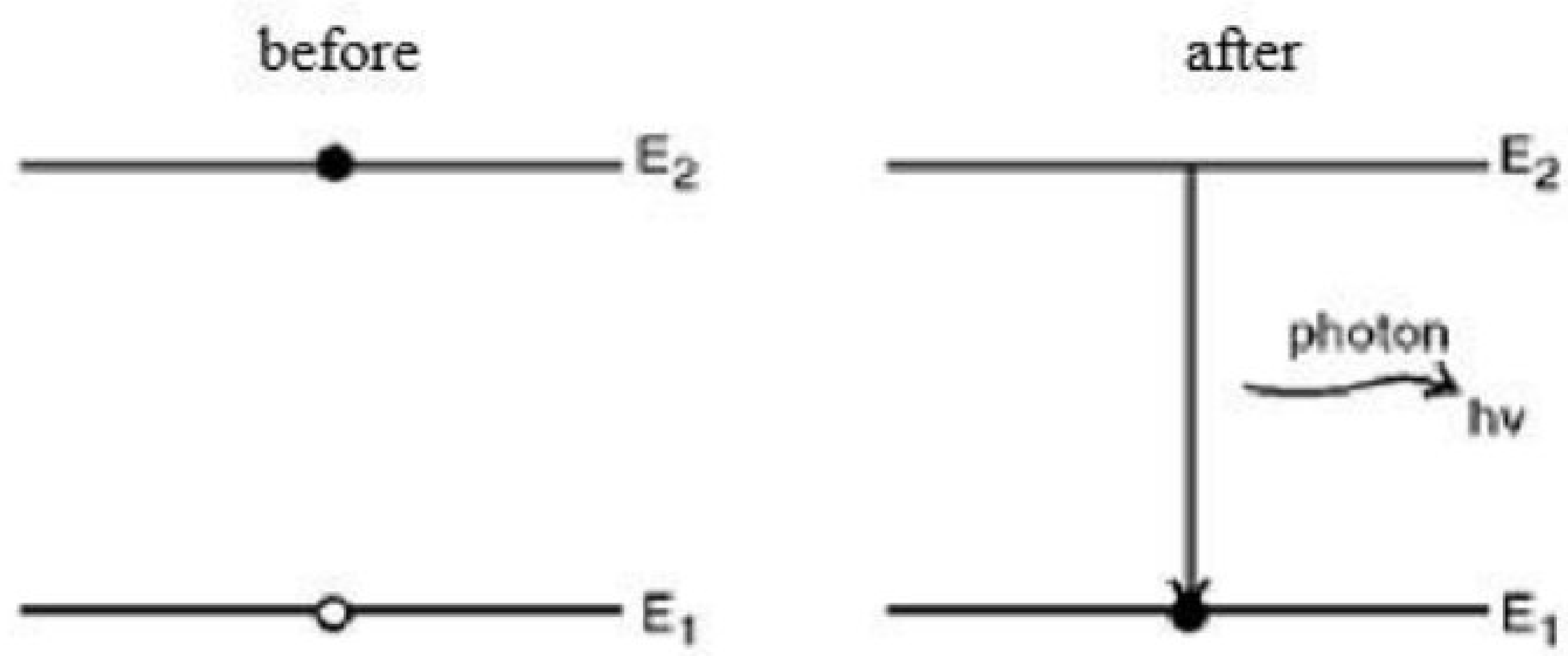
$$R_{abs} = B_{12} N_1 U(v)$$

Where B_{12} = Einstein's coefficient for absorption of radiation

N_1= Number of atoms present in lower energy state E_1

$U(v)$ = Energy density of the stimulated radiation

(2)Spontaneous Emission: When an atom in an excited state makes a transition into ground state by emission of a photon then it is called spontaneous emission. The radiation emitted by each atom has a random direction and random phase so emitted photons are incoherent.



Spontaneous emission

$$E_2 - E_1 = hv$$

h = Planck's constant
v = frequency of radiation

The process may be expressed as

$$A^* = A + hv$$

Where A is an atom in the lower energy state E_1 and A^* is an exited atom in higher energy state E_2

Let N_1 and N_2 number of atoms be the per unit volume in energy state $E_1$and E_2.

The number of spontaneous emission transition per unit time per unit volume or probability rate of spontaneous emission is given by

$$R_{sp} = A_{21}N_2$$

Where A_{21} = Einstein's coefficient for spontaneous emission which depend on properties of energy state $E_1$and E_2 . It is also called probability of spontaneous emission.

N_2= Number of atoms present in higher energy state E_2

(3)Stimulated Emission: When an atom in a higher energy state is forced by a photon to make a transition into ground state resulting into emission of another photon is called stimulated emission. The direction of propagation, energy, phase and state of polarization of the emitted photon is exactly same as that of incident stimulating photon. Two photons are in coherent.

Stimulated emission

$$E_2 - E_1 = hv$$

h = Planck's constant
v= frequency of radiation

The process may be expressed as

$$A^* + hv = A + 2hv$$

Where A is an atom in the lower energy state E_1 and A^* is an exited atom in higher energy state E_2.

Let N_1 and N_2 number of atoms be the per unit volume in energy state E_1 and E_2.

The number of stimulated emission transition per unit time per unit volume or probability rate of stimulated emission is given by

$$R_{st} = B_{21}N_2U(v)$$

Where B_{21} = Einstein's coefficient for stimulated emission of radiation

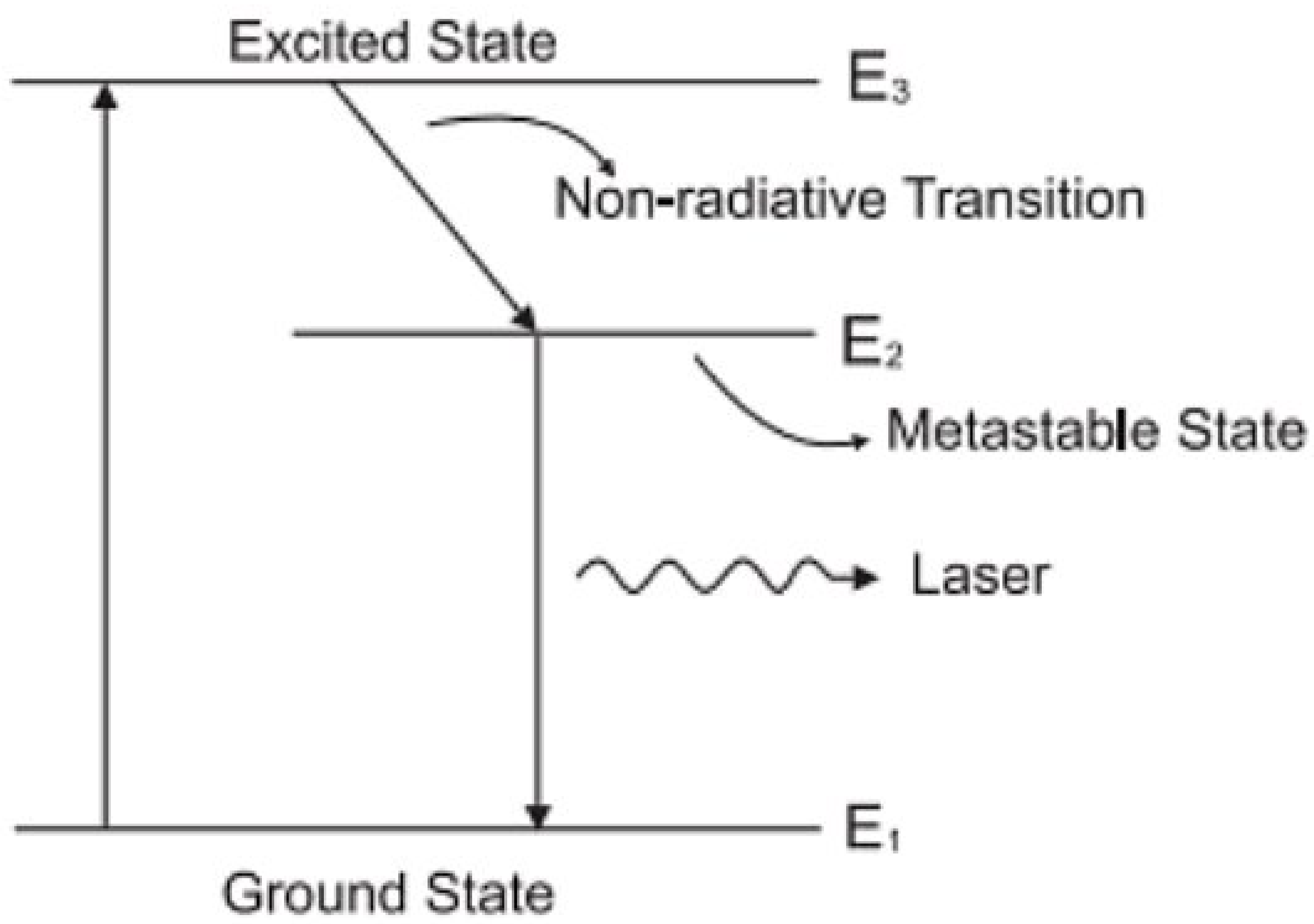
N_2 = Number of atoms present in higher energy state E_2

$U(v)$ = Energy density of the stimulated radiation

Metastable state: In an atomic system, atoms have discrete energy states. The atoms occupy the lowest energy state, known as the ground state. There are many higher energy states, some of higher energy states are metastable which having longer life time about 3×10^{-3}second.

(1) Upward transition from lower energy state to metastable state or downward transition from metastable to lower energy state is not readily possible. So spontaneous emission from metastable state is not possible.

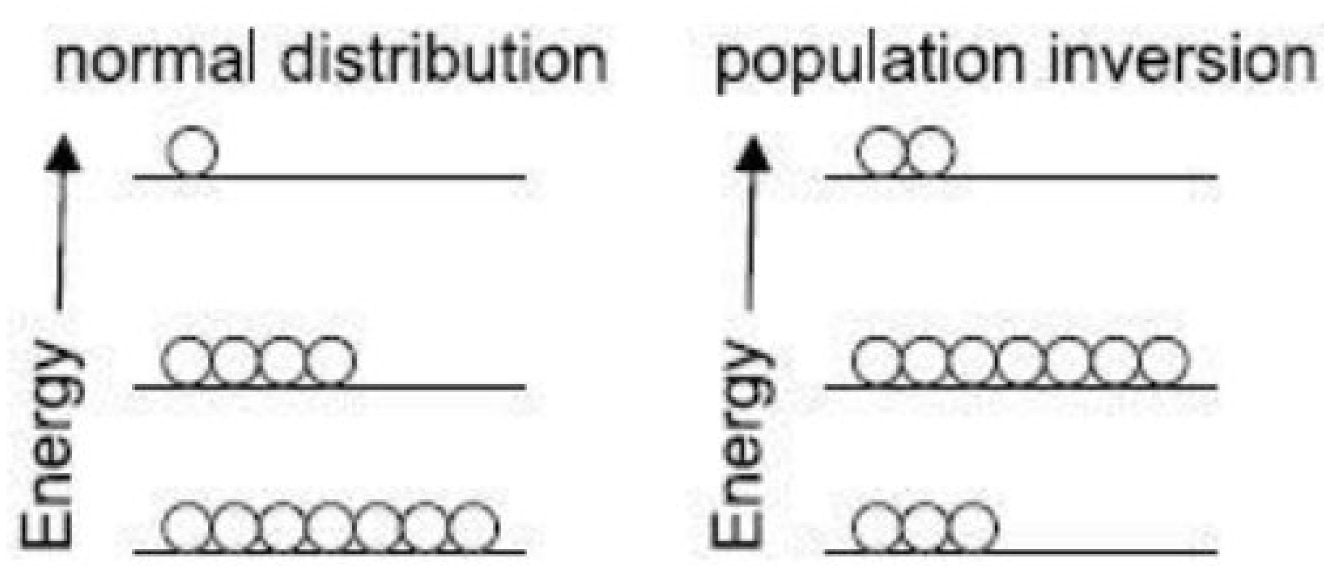
(2)When a photon of suitable energy collides an atom in a metastable state, then atom can drop to lower energy state by emitting an additional photon.



Population Inversion: Under ordinary conditions of thermal equilibrium, the number of atoms in higher energy state is considerably smaller than the number in lower energy state so there is very little stimulated emission compared with absorption. If however, by some means the number of atoms in higher energy state be made sufficiently larger than the number of atoms in the lower energy state, then stimulated emission is promoted. The state in which the number of atoms in the higher energy state exceeds that in the lower energy state is known as population Inversion.

Let N_1 and N_2 number of atoms be the per unit volume in energy state E_1 and E_2 .

For population inversion $N_2 > N_1$



Significance of population inversion in the operation of LASER:

(a)To increase the probability of stimulated emission, the number of atoms in the higher energy state must be greater than the number of atoms in the lower energy state. This is a precondition of LASER

(b)It makes LASER possible with the help of metastable state.

Pumping:

1. The process of raising large number of atoms from lower energy to a higher energy level is called pumping.

Types of pumping:

1. Optical pumping: which uses strong light source for excitation.
2. Electrical pumping: which uses electron impact for excitation.
3. Chemical pumping: which uses chemical reactions for excitation.
4. Direct pumping: which uses direct conversion of electric energy into light energy.

Differentiate between spontaneous and stimulated emission:

spontaneous	stimulated emission
1. When an atom in an excited state makes a transition into ground state by emission of photon, this process is called spontaneous emission.	1. When an atom in a higher energy state is forced by a photon to make a transition into ground state resulting into emission of another photon is called stimulated emission.
2. It is not useful for LASER.	2. It is useful for LASER.
3. It can not be controlled from outside.	3.It can be controlled from outside
4.The probability of spontaneous emission to take place is higher than the probability for stimulated emission to take place	4.The probability of stimulated emission to take place is higher than the probability for spontaneous emission to take place
5. Emitted photons are incoherent.	5. Emitted photons are coherent.
6.The net intensity is proportional to the net intensity is proportional to the number of radiating atoms $I=I_0N$	6.The net intensity is proportional to the net intensity is proportional to the square of number of radiating atoms $I=I_0N^2$
7.Emitted light is not monochromatic	7. Emitted light is of very high intensity.

Active medium: It is the material in which the laser action takes place. The active medium is an assembly of atoms, molecules or ions (in solid, liquid or gaseous form), which acts as an amplifier for light beams. The atoms which causes laser action, are called active centers. The rest of atoms of active medium acts as host and supports active centers. It may be solid crystals such as three level or Nd: YAG, liquid dyes, gases like CO2 or Helium - Neon, or semiconductors such as GaAs .This medium decides the wavelength of laser radiation. Active mediums contain atoms which can produce more stimulated emission than spontaneous emission and cause amplification they are called “Active Centers”.

Resonant cavity: Stimulated emission can be increased by increasing the radiation density. It can be done by resonant cavity.

An optical resonator knows as fabry perot resonator that consists of two mirror facing each other between which an active medium is placed. One of the mirrors is fully silvered and other is partially silvered .The mirror reflects most of the output energy back to the active medium. It provides positive feedback to the active medium. Laser beam come out from partially silvered mirror.

Role of resonant cavity:

(1) It is to provide a positive feedback of light into the lasing medium.

(2) To sustain laser action

(3)A chance spontaneously emitted photon by an excited atom acts as the input and induces stimulated emission.

Working: When laser is start, Photons emit spontaneously in all directions, few photons travel along the axis of cavity. Number of photons increases when passing through the active medium because these photons stimulate emission of more photons and decreases when reflected off the mirrors.

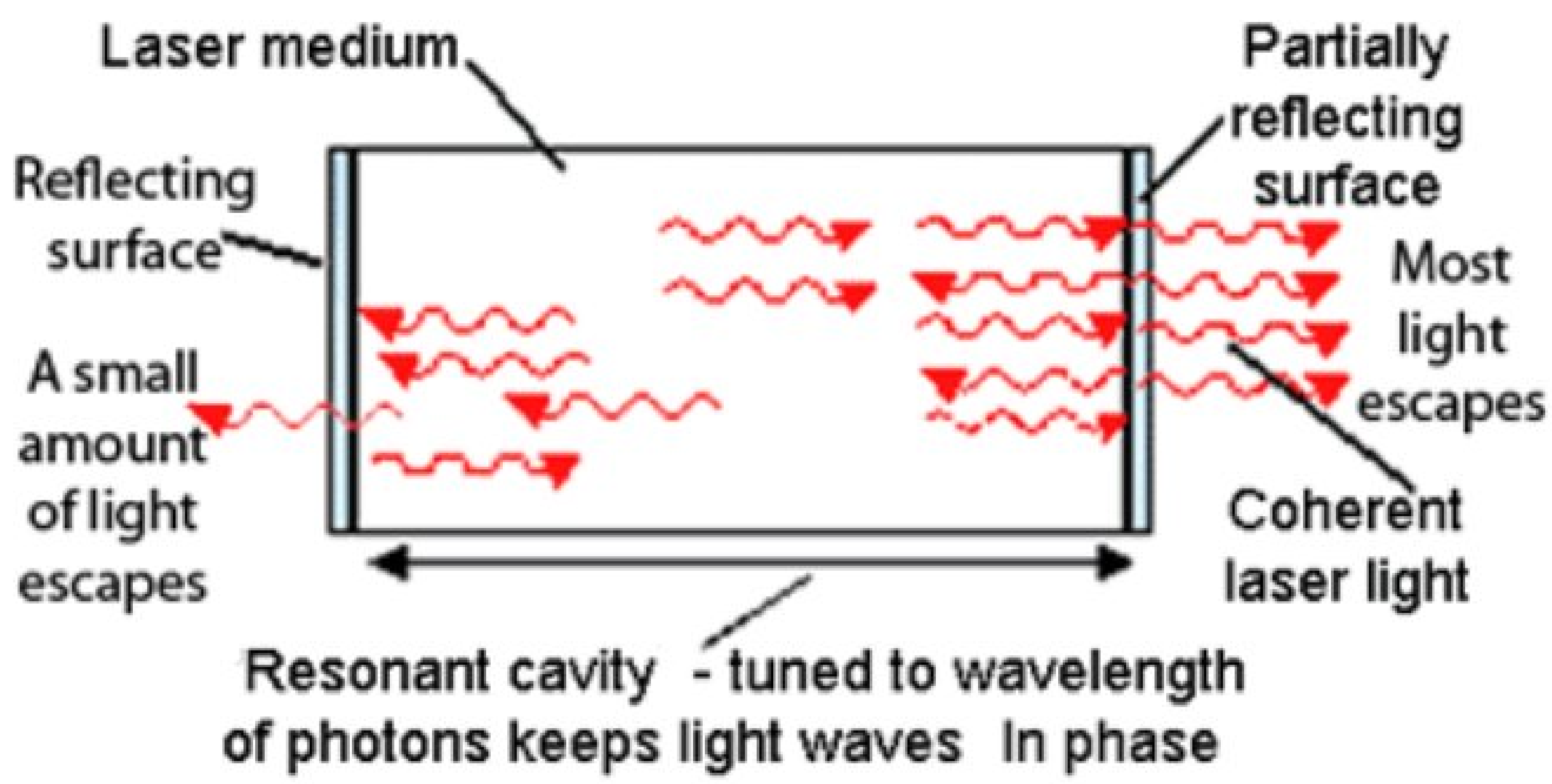
let a photon incident normally on one of the faces of the mirrors the ray suffer multiple reflections,bounching to and fro between the mirrors at each reflection the amplitude reduced by a factor $r_1 = \sqrt{R_1}$ and $r_2 = \sqrt{R_2}$

Where R_1 is reflectance of fully silvered mirror and R_2 is reflectance of partially silvered mirror. The wavelength at which resonances occur and standing waves are set up when

$$\lambda = \frac{2L}{m}$$

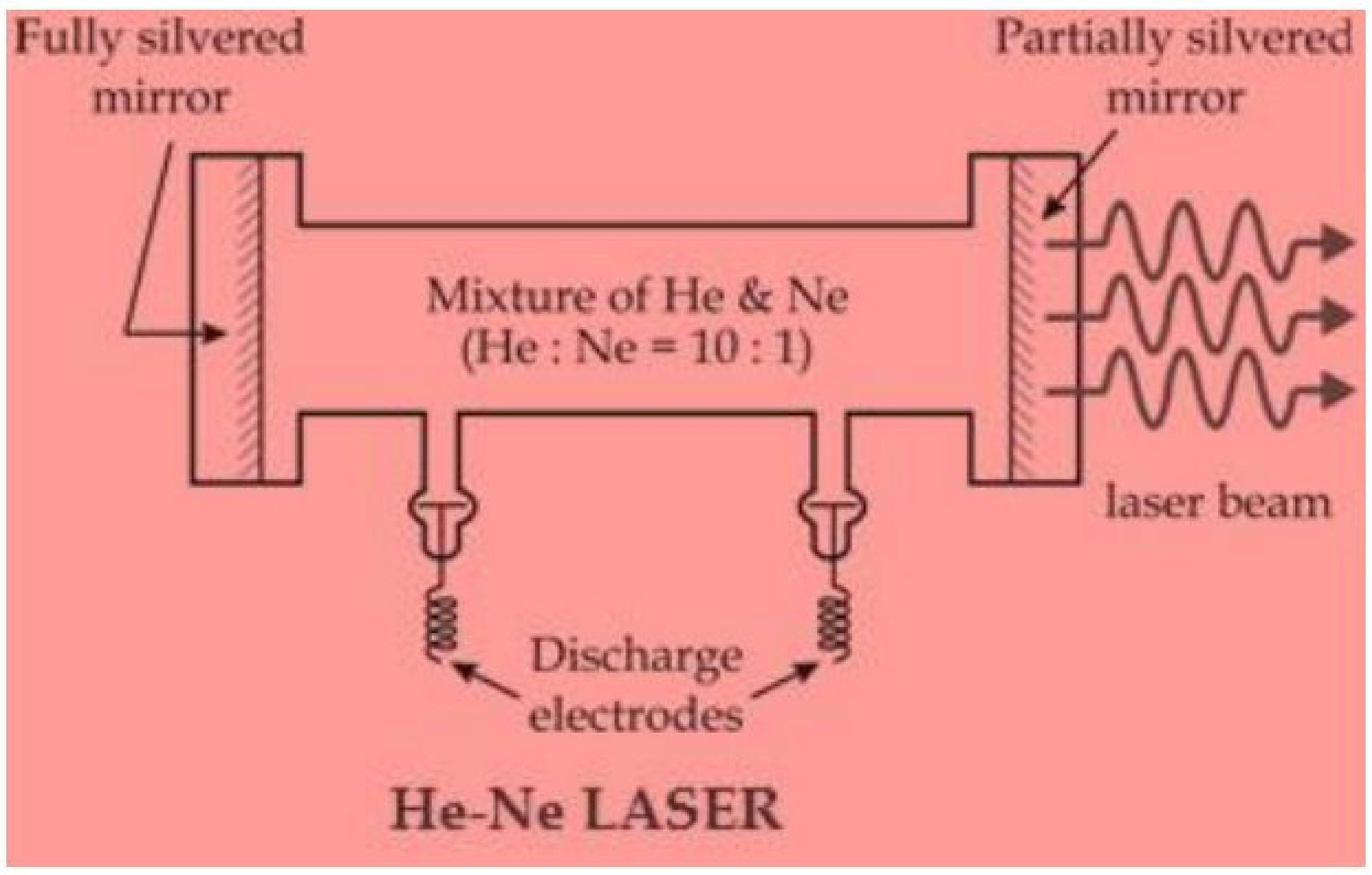
Where L distance between two mirrors,

m= mode number



Helium–neon (He–Ne) Laser:

Helium neon laser used a mixture of 10:1 (He:Ne) for its active medium. It consists of a long and narrow discharge tube of diameter of about 1 cm and about 80 cm long. The mixture is at a pressure of about 1 mm of Hg, the partial pressure of helium gas being 5 to 10 times that of neon



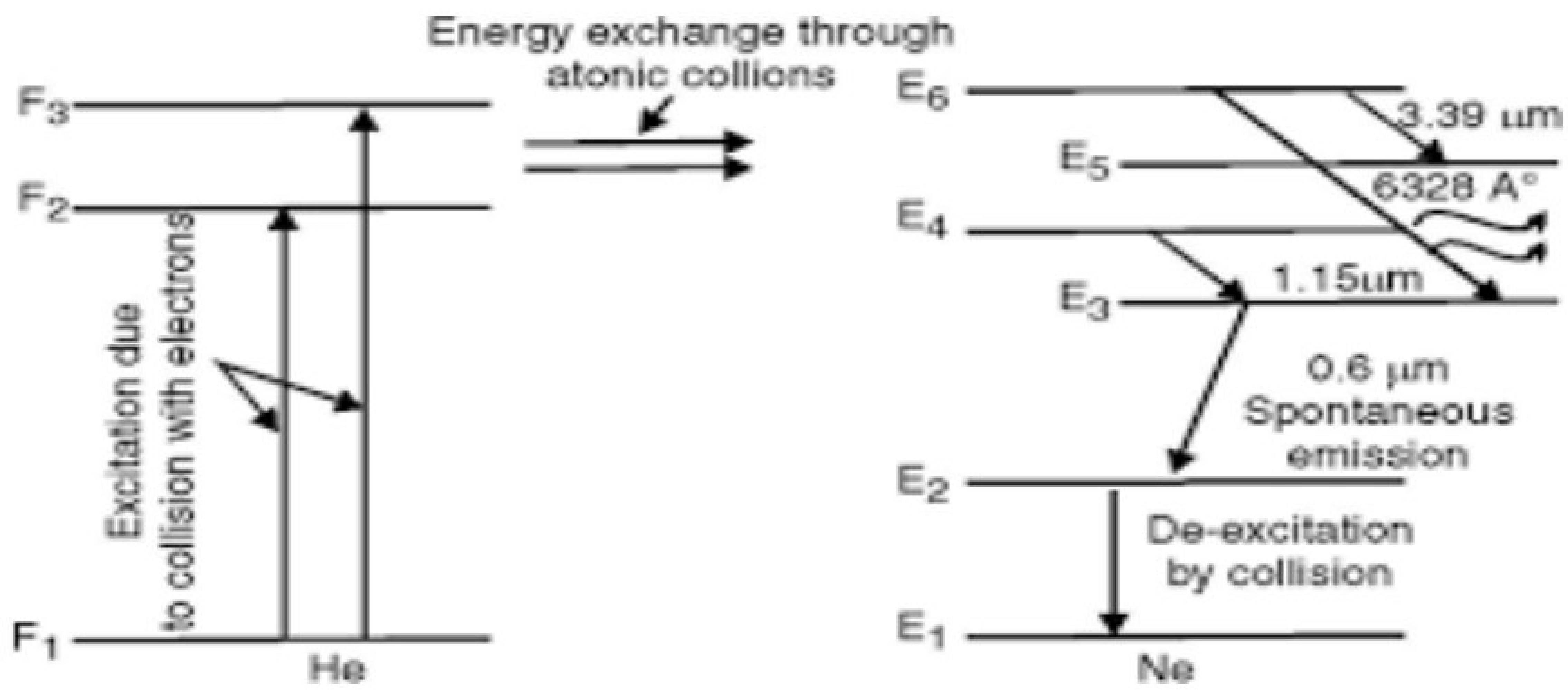
Construction:

Active medium: Ne atoms act as active centers and responsible for the laser action, while He atoms are used to help in the excitation process

Pumping Source: The pumping is done through electrical discharge by using electrodes that are connected to a high frequency alternating current source

Optical resonator system: The optical cavity of the laser usually consists of two concave mirrors or one plane and one concave mirror: one having very high (typically 99.9%) reflectance, and the output coupler mirror allowing approximately 1% transmission

Working and transition diagram of Helium Neon Laser



1. Pumping of He atoms: When electric discharge is passed through the gas mixture of He and Ne, electrons are accelerated down the discharge tube in which mixture of He-Ne is placed. These accelerated electrons collide with helium atoms and excite them to higher energy levels (let us say F_2 and F_3). These levels happen to be Metastable and thus the He atoms spend a sufficient amount of time there before getting de-excited

2. Achievement of population inversion of Neon atoms: Some of the excited states of Ne atoms correspond approximately to the same energy of the excited levels F_2 and F_3 of He. Thus, when He atoms in levels F_2 and F_3 collide with the Ne atoms in the ground state E_1 , then energy exchange takes place and this results in the excitation of Ne atoms to the levels E_4 and E_6 and de-excitation of the He atoms to the ground level F_1 . As the helium atoms have longer life time in excited states F_2 and F_3 thus this process of energy transfer has high probability .Therefore, the electric discharge through the gas mixture continuously populates the Ne excited levels E_4 and E_6. This helps to create a state of population inversion between the levels E_6 (or E_4) and lower energy levels E_5 and E_3. Therefore the purpose of He atoms is to help in achieving a population inversion in the Ne atoms

3. Achievement of laser: The following three transitions will occur:

E_6 to E_5 with laser wavelength of 3.39 μm or 33900 Angstroms.

E_6 to E_3 with laser wavelength of 6328 Angstroms.

E_4 to E_3 with laser wavelength of 1.15 μm or 11500 Angstroms.

The wavelengths of 3.39 μm and 1.15 μm corresponds to infrared region and wavelength 6328 Angstroms corresponds to red light wavelength (visible region).

4. The excited Ne atoms drop down from levels E_3 to E_2 through spontaneous emission and this process will emit a photon of wavelength 0.6 μm. As the level E_2 is also Metastable, there is a probability of excitation of Ne atoms from E_2 to E_3 leading to quenching of the population inversion. To eliminate quenching, the narrow discharge tube is used because Ne atoms de-excited to level E_1 from E_2 through collisions with the walls of the tube.

5. Mirrors of the optical resonators are so designed to show low reflectivity for wavelengths 3.39 μm and1.15 μm. Thus photons of these wavelengths will be eliminated. Therefore, the photons of wavelengths 6328 Angstroms will move back and forth between the end mirrors, these waves will stimulate emission of the same frequency from other excited neon atoms, and the initial wave with the stimulated wave will travel parallel to the axis. Thus laser of wavelength 6328 Angstroms emerges through the partially reflected mirror.

Merits:

1. Continuous output laser source
2. Highly stable
3. No separate cooling is required

Demerits:

1. Low efficiency and low power output
2. Gases are novel medium for laser as gases are found in the purest form so their optical properties are well defined.

Application: It is used to read bar code, in scanner, and in printers and in holography etc.

Role of helium atoms: Helium atom help in achieving a population inversion in the Neon atom.

The lighter helium atoms can be easily pumped up to their excited state, the much heavier atoms could not be raised efficiently without collision with helium atom.

Short Question answer:

Q1.Which Laser can be used to generate pulse laser output?

Ans: Nd : YAG laser Neodymium doped yttrium aluminium garnet laser

Q2. What is the method of achieving population inversion in Semiconductor diode laser?

Ans: Achievement of population inversion: When p-n junction diode is forward biased, then there will be injection of electrons into the conduction band along n-side and production of more holes in valence band along p-side of the junction. Thus, there will be more number of electrons in conduction band comparable to valence band, so population inversion is achieved.

Q3. Why the diameter of the He is: Ne LASER tube kept small.

Ans: In He- Ne laser, Ne atom has E_1, E_2, E_3, E_4, E_5 and E_6 states, population inversion is achieved between the levels E_6 and E_3 which causes laser beam. As the level E_2 is also Metastable, there is a probability of excitation of Ne atoms from E_2 to E_3 leading to quenching of the population inversion. To eliminate quenching, the narrow discharge tube is used because Ne atoms de-excited to level E_1 from E_2 through collisions with the walls of the tube (atomic collision with tube wall increases).

Q4. What are the factors on which stimulated Emission of radiation in Laser depends upon?

Ans: Stimulated Emission of radiation depends on

(1) Number of atoms per unit volume present higher energy state.

(2) Energy density of the stimulated radiation.

Questions:

Short Answer Question:

(1) What is population inversion state? Explain its significance in the operation of LASER.
(2) What is an active medium in LASER?
(3) State all the unique properties of LASER?

Long Answer Question:

(1) Explain the terms Spontaneous emission, Stimulated emission, metastable state, Population inversion and pumping.
(2) Describe the construction and working of He-Ne laser. What are its merits and demerits? What is the role of helium atoms?
(3) Describe the construction and working of Nd-YAG laser.
(4) Explain the construction and reconstruction of hologram

MCQ:

1. Directional property of laser can be used in

(a) Surveying (b) remote sensing (c) Lidar (d) All correct

Ans: (d)

2. Nd - YAG laser is a

(a) Two level laser (b) three level laser (c) four level laser (d) five level laser

Ans: (c)

3. In Nd-YAG Laser YAG Means

(a) Yttrium Aluminum Garnet (b) $Y_3Al_5O_{12}$ (c) Indium aluminum garnet

(d) Both A and B

Ans: (d)

4. The pumping source in Nd:YAG laser is

(a) Chemical (b) optical (c) electrical (d) mechanical

Ans: (b)

5. The active medium in Nd: YAG laser is

(a) Nd (b) YAG (c) Y (D) AG

Ans: (a)

6. Example of solid state laser is

(a) He-Ne Laser (b) CO_2 laser (c) Nd:YAG (d) dye laser

Ans: (c)

7. In which region, laser emission occurs in Nd: YAG laser

(a) IR region at 1.06μm (b) visible region (c) UV region (d) RF region

Ans: (a)

8. The ratio of He to Ne in He-Ne laser is

(a) 1:10 (b) 2:14 (c) 3:11 (d) 10:1

Ans: (d)

9. The active medium in He-Ne laser is

(a) He (b) Ne (c) He-Ne (d) none

Ans: (b)

10. The role of He in He-Ne laser is

(a) He is an active medium (b) population inversion takes place in He

(c) Stimulated emission takes place in He (d) He atoms help in exciting Ne atoms

Ans: (d)

11. The reason for narrow tube in He-Ne laser

(a) atomic collision with tube wall increases (b) atomic collision with tube wall decreases

(c) atomic collision with tube wall constant (d) None

Ans: (a)

12. Population inversion in laser means

(a) number of atoms in ground state are more than number of atoms in excited state

(b) number of atoms in ground state are less than number of atoms in excited state

(c) number of atoms in ground state is equal to number of atoms in excited state

(d) None

Ans: (b)

13. Metastable state has life time approximately

(a) 10^{-9} (b) 10^{-3} (c) 10^{-10} (d) 10^{-12}

Ans: (b)

14. An excited state (except metastable state) has life time about

(a) 10^{-8} (b) 10^{-3} (c) 10^{-10} (d) 10^{-12}

Ans: (a)

15. He-Ne laser is a

(a) Two level laser (b) three level laser (c) four level laser (d) five level laser

Ans: (c)

16. The pumping source in He-Ne laser is

(a) chemical (b) optical (c) electrical discharge (d) mechanical

Ans: (c)

17. Hologram is the result of

(a) Interference of object and reference beam

(b) Polarization of object and reference beam

(c) Diffraction of object and reference beam

(d) Both Polarization and diffraction of object and reference beam

Ans: (a)

18. Each part of hologram contains information about

(a) Entire object (b) particular part of the object (c) important part of the object

(d) Front side of the object

Ans: (a)

19. The information carrying capacity of a hologram is

(a) more than an ordinary photograph (b) less than an ordinary photograph

(c) equal to an ordinary photograph (d) All option correct

Ans: (a)

20.In holography, which of the following optical phenomena are involved?

(a) interference, diffraction (b) polarization, diffraction

(c) interference, refraction (d) reflection, diffraction

Ans: (a)

21. In holography

(a) Only phase information is recorded (b) only amplitude information is recorded

(C) Both phase and amplitude get recorded (d) neither phase nor amplitude gets recorded

Ans: (c)

22. What is the region enclosed by the optical cavity called?

(a) Optical Region (b) Optical System (c) Optical box (d) Optical Resonator

Ans: (d)

23. Which of the following is not a property of emitted light in stimulated emission?

(a) incoherent (b) unidirectional (c) monochromatic (d) high intensity

Ans: (a)